$$\frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE}(0)}}:\ \kappa \text{ cancels, no galaxy knob left}$$

the key: $\rho_{\rm DE}(z)$,
measured by baryon acoustic
oscillations and supernovae

nobody can turn this one